

THE AGE

MELBOURNE LIVING

edited by STEPHEN DOWNES and PETER WEINIGER (Epicure)

Pictures: JOHN WOODSTRA



But four young Oysters hurried up,
All eager for the treat:
Their coats were brushed, their faces washed,
Their shoes were clean and neat—
And this was odd, because, you know,
They hadn't any feet.

From 'Through the Looking-Glass' by Lewis Carroll,
Illustrated by Sir John Tenniel.

THE OYSTER REVOLUTION

TREVOR DIX forms his hands into a scoop and dredges deeply into what looks like coarse river sand. An argument follows. Half a million? One hundred thousand? Estimates replace guesstimates and a figure is decided on. He holds 180,000 oysters in his hands.

In less than 18 months, almost all of them — and millions of their siblings — will be at full-size and on the market, the climax of a revolution in the Australian oyster industry.

In fact it is more than that. Australian mariculture — as in agriculture, but of the sea — which up to now has had mostly the attributes of a cottage industry, is about to enter the 20th Century. Millions of dollars have been invested in this dramatic turnabout in Australian oyster growing and marketing, and the rewards will be huge.

They will go to Cheetham Limited, which traditionally harvested salt but has lately diversified into garden products and animal feed supplements. By the financial year 1986-87 it could be producing about two-thirds of the 150 million oysters presently consumed each year in Australia. (Consumption is expected to increase when Cheetham enters the market.) And it will have taken only three years to get there.

Next calendar year Cheetham hopes to sell 20 million oysters at a profit of about \$2.5 million. If it reaches, soon after, its 100 million target, it will be looking at annual oyster profits of \$12.5 million. No one I have spoken to can remember marine cultivation in Australia of such a scale.

The company plans to market its product in a variety of ways: closed, opened and washed, and eventually — via a new processing plant in Geelong — frozen and packaged. Oysters kilpatrick or mornay at hand in the freezer are not, if all goes well, too many rushed dinner parties away.

TO APPRECIATE how big an effect Cheetham's incursion into oyster growing will have on the industry you might like to think of the differences between a sophisticated radio-controlled model aeroplane and a kite.

The Sydney rock oyster (*Crassostrea commercialis*) industry, which has dominated the market up to now, is like a kite. It needs wind to fly. In oyster-growing terms this means producers rely on natural fertilization, then the catching of microscopic oyster larvae on sticks suspended in the temperate waters of estuaries.

EXCLUSIVE

By STEPHEN DOWNES

It is hit and miss, and in the three years the Sydney rock oyster takes to reach edible size, excessive heat, amoebic parasites, mud-worm and other predators can decimate the crop. For the trouble he went to in coaxing his oysters to adulthood, the grower then must prise the crop off the sticks. Like a lot of farmers, oystermen are elated if they get two good years in five and half their crops to the table.

But other problems also dog the New South Wales industry. Health scares led to new purification rules in 1983 that require Sydney oysters to spend 48 hours in sea water irradiated with ultra-violet light, which kills bacteria. Traditionally, too, they have been usually taken to market opened after having been washed in fresh water, a treatment true oyster connoisseurs abhor. Bill Cook, Cheetham's managing director, summarises the rock oyster industry thus: "They're in the Stone Age up there. You ought to see it. It's terrible, unbelievable."

Compare this with Cheetham's model aeroplane. In fact, perhaps a better comparison between the industries might be kite to 747 jetliner.

WE ARE bouncing in a mini-bus along an embankment, one of a regular pattern of welts across vast salt pans north of Adelaide. On either side are saltwater ponds and mangrove flats. Myriad flocks of birds have invaded here — perhaps lived here forever; pelicans, cormorants, gannets. A hundred species have been logged.

The land is ICI's (the company is a big salt producer for its chemical processes) and at a junction of three dikes is a corrugated-cement shed about the size of a country church-hall. It is an oyster hatchery, built during the '70s but always a Cinderella until Cheetham came along.

Now it has "Mermaid Oysters" in blue lettering on a plinth slapped to it and the blue mermaid familiar from salt packets as an added decoration.

Cheetham had been experimenting in its salt ponds at Lara, near Geelong, with Pacific oysters from 1982; it had also bought spat, baby oysters, until the end of 1983, from ICI. Then ICI lost interest in oysters in a rationalisation of the company. In January last year, Cheetham began leasing the hatchery with an option to buy.

Overall, it has sunk \$5 million into oyster growing, adapting re-

search done in Europe, Japan and the United States. The oyster it favors, too, is surprisingly amenable to cultivation, grows to table size in half the time of a Sydney rock oyster and is excellent eating.

The Pacific oyster (*Crassostrea gigas*) is farmed throughout Europe — mainly France — and Japan and Korea. Two shipments of it were brought by the CSIRO from Japan in 1947 and 1948 and "planted out" in Tasmania and Western Australia. Descendants of these oysters form the present Tasmanian industry, which supplies Pacific oysters to a handful of discriminating Melbourne restaurants.

Very soon though, they will be available in vast numbers to the southern mainland from Lara and South Australia.

Inside the hatchery, our delegation — Bill Cook and a handful of Cheetham people, a couple of representatives of the company's biggest investors, photographer John Woodstra and I — is met with a vision of studied cleanliness. Three men dressed in spotless white boiler suits and wearing white gumboots run the operation.

There are three big cylindrical tanks the size of backyard swimming pools and a series of small rectangular ones in which are suspended fine circular sieves containing growing oysters. Green saltwater from the ponds surges through all these containers, pumped and slurped through a network of crisp grey and blue and red plastic pipes and valves.

But it all starts in a shallow, small, rectangular bath where up to about 40 breeding oysters are placed. The temperature of the water is increased until the oysters begin to ejaculate sperm and eggs.

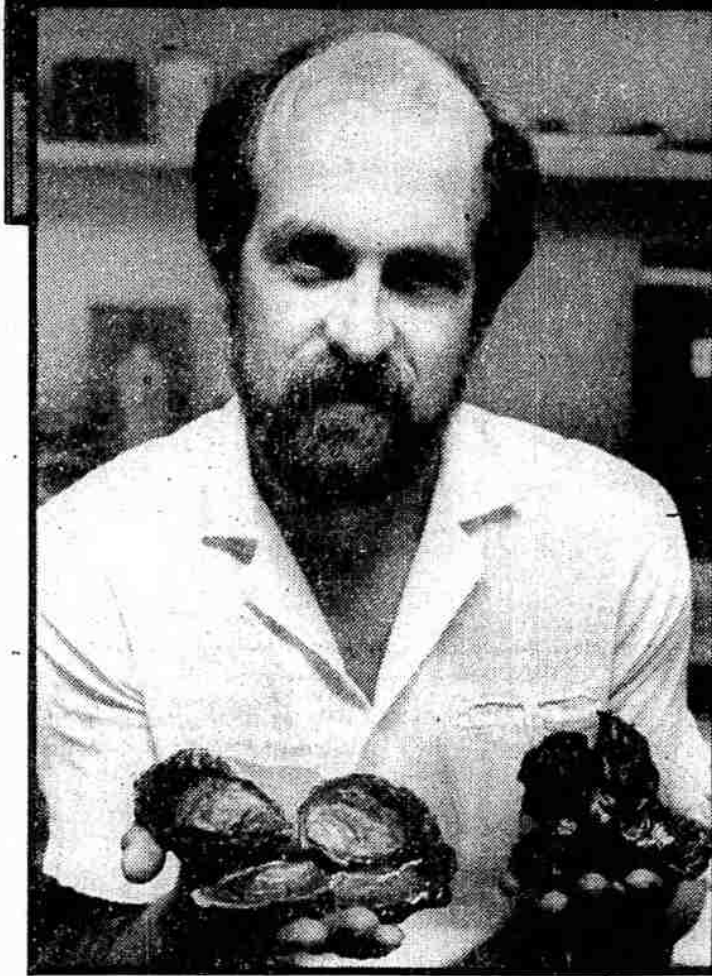
Male and female oysters are put into separate buckets half-filled with sea water and allowed to finish their secretions. Water containing the sperm and eggs are then mixed and a resultant bucketful can contain about 50 million fertilised eggs, which are then tipped into one of the swimming pools. For about 12 to 13 days, the free-swimming oyster larvae, which look like tiny Ds under the microscope, are fed thrice daily with special strains of algae imported from the United States, Tahiti and the United Kingdom.

The algae grow in huge plastic bags of saltwater in a temperature-controlled hygienic room within the hatchery. But the real quantum leap in technology comes next. Oysters must settle and stick to something if they are to live happy lives. In open New South Wales waters they settle on the oyster growers' sticks — sometimes.

In the Adelaide hatchery, the larvae settle on ground-up scallop shell which is the consistency of



Peter Rankin: Cheetham's revolution makes oyster-growing easy.



Neil Hickman holds local oysters (left) and Cheetham's Pacifics.

fine sand. From that point on, the grower is effectively handling discrete oysters with all the attendant advantages that that entails.

Once the larvae have settled, they grow through various stages on the sieves until, at about eight to 10-weeks-old and the size of sequins, they are sent — damp and in a plum-pudding-shaped mesh — in insulated containers to Lara in Victoria and Price, which is on the shore of St Vincent's Gulf about 90 kilometres north-west of Adelaide. About 3½ million oysters occupy a litre.

Cheetham has plans for two more hatcheries, one at Lara and the other in Tasmania, where it has a half-share in the Tasmanian Shellfish Company, to which Trevor Dix, one of the country's expert mariculturists, is the consultant. The advantage of having a Tasmanian arm to the operation is

enormous. Mainland-growing Pacific oysters become flaccid and transparent in summer heat after spawning; in Tasmania they grow in deeper, cooler water and retain their condition.

BILL COOK says there can't be a worse place in the world to try to harvest salt than Geelong. Yet the more the company moves into oysters the more it realises that evaporation ponds under overcast skies are a godsend.

At Lara, the canniness of its operation is obvious. Peter Rankin, Cheetham's production manager in the mariculture division, sweeps his arms over hundreds of hectares of ponds. "We've got 1.2 million oysters growing here at the moment. We could do 10 times that many."

Continued Living 4

Bay oyster promises much

IN THE times in which we live, it is proper to see nothing unusual in wanting to change your sex.

The native southern Australian oyster, which is prevalent in Port Phillip and Westernport bays, has been broad-minded about such things for years; it actually practises transsexuals, sometimes at the rate of a change every three weeks. It can also do a classic hotel/motel, AC/DC and be a bit of both sexes.

Unlike the Pacific oyster, native oysters (*Ostrea angasi*) are fertilised and incubated within the mother's shell. And compared with the rock-dwelling species (such as the Pacific) they are rounder and flatter. They are also tawny-colored; rock oysters are grey.

But most importantly for gastronomes, they are good to eat, subtly different from the Sydney rock and Pacific varieties. Harvested at table size, they are sweet but not so full as samples of Cheetham's Pacific oysters that had been grown in fertilised saltwater ponds. Their taste, however, is excellent and subtly different from the Pacific or Sydney rock.

If marine biologist Neil Hickman's studies at the Queenscliff Marine Science Laboratories are conclusive, oyster farmers will in the next few years be producing *Ostrea angasi* by the millions in Port Phillip Bay, and Victorian diners might one day be able to order seafood platters compris-

ing three different oyster varieties. "All preliminary work has been very encouraging," he says.

Mr Hickman and his three assistants began their three-year investigation into the feasibility of farming the native oyster only in January after some establishing work late last year. Funded by commercial fishermen and the Federal Government through the Fishing Industry Research Trust, 35 exploratory licences have been issued to grow the species, 20 in the bay and the rest spread along the Victorian coast. Since publicity of the research, prospective oyster-farmers have been "beating the doors down" to get licences for trial plots.

Mr Hickman (who with a technical officer is the State Government's contribution to the project) is charged with ensuring that there is a "sound biological basis for the industry to develop", as he puts it.

Almost fished out by about 1900 after decades of dredging, the native oyster is once again prolific in the bay. According to Neil Hickman its growth rates have been "very encouraging". At median maturity it is saucer-sized, but it will be harvested smaller for eating.

Aborigines ate them, and some Mornington Peninsula families have for generations used the mature "mud" oysters, as the *angasi* have been known, to make soup. A hundred years ago they were also burned in piles on bay beaches to produce lime that went into making mortar.

Mr Hickman says their gastronomic potential these days "seems enormous".

At Queenscliff you see ponds of surging saltwater, plastic pipes and valves and microscopic examination of larvae and spat, just as in the Cheetham operation. But Mr Hickman is only just beginning to set up his hatchery and he is, of course, behind Cheetham in funding terms.

The MSL itself is conducting research at six sites around the Victorian coast where the oysters are being tested for their maricultural suitability. There is a lot of oyster larvae in the bay and the best spots for catching it — on things such as plastic and cement sheet and scallop shells — are being determined. Once the larvae have settled on something solid and manageable, the best places and methods for growing them have also to be worked out.

Media reports so far have concentrated on the alleged competition between the Pacific and native oysters and the organisations backing their development. But both Neil Hickman at the MSL and Peter Rankin, for instance, at Cheetham, deny it exists. In fact, they often consult one another. They both say their work will conclude in giving consumers a greater range of choice.

Both the MSL and Cheetham are working towards intensive mariculture, more jobs and, for lovers of seafood, fresher and better oysters.

A Chinese solution to train traumas

FRANKLY I haven't forgiven the Australian Railways Union for staging a strike during Cup Week.

Cup Week is sacred in Melbourne, more important than Christmas, Good Friday or even the Grand Final.

That's it; persuade visitors from all over the globe to come to town, then leave them stranded. The battle out to Flemington was a desperate affair, with that final struggle to the barrier, fighting one's way through a mud-laden car park.

There is a fine old tradition in Australia for waging strikes at times designed to inflict the maximum torture on the public.

It used to be *de rigueur* to have a brewery strike immediately before Christmas, just when precious bottle supplies were running short. Then we have had communications strikes just when we have wanted to tune in to a vital Test match; petrol strikes or air strikes just when we are about to set out on holidays.

But what are we to do about the poor, suffering, inefficient railwayways? We create world records for late-running trains. We cancel up to 40 a day.

Melbourne's Metropolitan Transit Authority has just doubled its year's deficit, pushing it up to \$219 million. The level of its borrowings has almost doubled to \$1031 million.

And seeing that we have gone into the business of leasing our trains, what with high interest rates, it is all escalating. The debt and the deficit next year will be magnificent.

Can we really afford the railways any more? Now I have an idea that could save us all from ruin. It is something the Chinese discovered 50 years ago. It is cheaper, more efficient and healthier to put the public on bicycles.

It has been proved countless times that in any metropolitan area the bicycle is faster than public transport. What with the agony of waiting on platforms or trying to park a pollution machine, from point A to point B a bicycle will get you there quicker.

Now let us presume there are two million bicycle-sized citizens in Melbourne. Rather than run a railway system it would be cheaper to give them all a bicycle free. Not one of your cheap Taiwan affairs but a magnificent machine.



Divide two million into our annual MTA deficit and that comes to \$109.50 cents a head. Or look at our borrowings, \$1031 million. That comes to \$515.50 a head. Instead of spending all that money on something we don't like — the citizens seem to spend most of the time slashing the seats — why not give the money to the public.

Already you are saying there are the limp and the old who couldn't ride a bicycle. OK, we would have so much money left over that for special cases we could provide taxis.

Yes, there will be lots of money. Just at a quick estimate I reckon we'll make five billion by selling up railway assets. Flinders Street

Railway Station, with its elaborate dome, would make a lovely temple. The Seventh Day Adventists or the Hare Krishnas would find it irresistible.

In California there's a craze for eating in railway stations. There's a big chain of restaurants called Victoria Station. They are all modelled exactly on London's Victoria Station. They have clocks, platforms, everything and, fair dinkum, the customers eat in railway carriages. They love it.

We could do that. Ascot Vale, Moonee Ponds, Clifton Hill, all warm, friendly, throbbing with people eating, having fun pretending they are clunking along in trains. Instead of grumbling, growling, seat-slashing and complaining that the bloody things are late.

We have a series of lovely underground railway stations, each with an individual theme. For example, there's Parliament Station, which is just like Parliament with fake Corinthian columns and all. It could be a night club or a fascinating kinky massage parlor.

Peking gets in for underground department stores on the theory that they are better during nuclear attack. I am sure that Woolies and Myer would be interested in taking several of these.

Those enormous, wasteful railway yards we can turn into parks, cricket fields, bird sanctuaries.

And what do you think about the thousands of kilometres of railway line? Admittedly we could have bicycle paths, but there's room for more than that. I would sell off chunks for tennis courts, bowling rinks, golf courses, and for glorious housing schemes.

Some of these schemes we could name nostalgically after trains. We could have "The Red Rattler", "The Blue Harris" or, pardon my smile, "The Spirit of Progress". The apartments could have those test-your-strength sliding doors.

Under this system we would need only 10 per cent of the automobiles we have now. The air would be clean again. We would stop our annual slaughter and maiming of thousands on the roads. State treasurers would have money again beyond their dreams.

It would even be a pleasure once more to visit the city or the race-track.

The Minister for Transport, no doubt, is terribly busy talking to his insurers or his unions but if he cares to call me I'm very happy to discuss the idea in depth.

— Keith Dunstan

Special Gifts for Special Cooks

Magimix
the original Food Processor from France \$179.50

Rolling Pin \$12.50

Marble Pastry Board (and use it for CHEESE & FRUIT at dinner time) \$39.95

Stainless Steel Flour Sifter \$10.50

Recipe holder \$7.95

Pudding Cloth \$5.25 pair

Demonstrations:

CITY:	14th NOV.	11 - 3.00 PM
FRIDAY	15th NOV.	11 - 3.00 PM
CAMPBELL:	21st NOV.	11 - 3.00 PM
FRIDAY	22nd NOV.	11 - 3.00 PM
CARLTON:	28th NOV.	11 - 3.00 PM
FRIDAY	29th NOV.	11 - 3.00 PM

(CHARGES ... TO BE ADVISED)

House.

CITY: 271 LYON STREET 347 9244

CAMPBELL: THE JUNCTION (WITH COUNTRY ROAD) 82 2274

CITY: THE GALLERIA STATE BANK CENTRE ELIZABETH STREET 67 6096

CHADSTONE: Opening NOVEMBER